concours papers at https://www.meetlearn.com/ML GCE materials at https://www.meetlearn.com

# CAMEROON GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

## 0765Maths With Mechanics 1

**JUNE 2020** **ADVANCED LEVEL**

| | |
|---|---|
| **Centre Number** | |
| **Centre Name** | |
| **Candidate Identification No.** | |
| **Candidate Name** | |

Meetlearn.com

**Mobile phones are NOT allowed in the examination room.**

## MULTIPLE CHOICE QUESTION PAPER

**One and a half hours**

### *INSTRUCTIONS TO CANDIDATES*

***Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.***

1. USE A SOFT HB PENCIL THROUGHOUT THE EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

***Before the examination begins:***

3. Check that this question booklet is headed **Advanced Level– 0765 Math with Mechanics 1.**
4. Fill in the information required in the spaces above.
5. Fill in the information required in the spaces provided on the answer sheet using your HB pencil: **Candidate Name, Exam Session, Subject Code and Candidate Identification Number.** Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.
6. ***Answer All questions.***
7. **Mathematical tables (Formulae booklets) and calculators are allowed.**
8. Each question has FOUR suggested answers: **A, B, C** and **D**. Decide which answer is appropriate. Find the number of the question on the Answer Sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen.
   For example, if **C** is your correct answer, mark **C** as shown below:
   [A] [B] ~~[C]~~ [D]
9. Mark only one answer for each question. If you mark more than one answer, you will score a zero for that question. If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
10. Avoid spending too much time on any one question. If you find a question difficult, move on to the next question. You can come back to this question later.
11. Do all rough work in this booklet using the blank spaces in the question booklet.
12. **At the end of the examination, the invigilator shall collect the answer sheet first then the question booklet. DO NOT ATTEMPT TO LEAVE THE EXAMINATION HALL WITH IT.**

concours papers at https://www.meetlearn.com/ML Buy Cameroon Textbooks at https://www.meetlearn.com/ML





©2020CGCEB

## SECTION A: PURE MATHEMATICS

1. $\frac{5x+1}{(x-1)(x+2)} \equiv$

A $\frac{2}{(x-1)} + \frac{2}{(x+2)}$

B $\frac{2}{(x-1)} - \frac{3}{(x+2)}$

C $\frac{2}{(x-1)} + \frac{3}{(x+2)}$

D $\frac{2}{(x-1)} - \frac{3}{(x+2)}$

2. The equations of the vertical asymptotes to the graph of $f(x) = \frac{4}{x^2-1}$ has

A $x = 1, x = 4$

B $x = -1, x = 4$

C $x = 1, x = -1$

D $x = -1, x = -4$

3. Two functions $f$ and $g$ are defined by $f(x) = 2x + 3$ and $g(x) = x - 2$. The composite function $g \circ f =$

A $2x + 1$

B $2x - 1$

C $2x + 6$

D $3x + 1$

4. If $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, then the cofactor of the element 6 is

A $-\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$

B $\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$

C $-\begin{vmatrix} -1 & -2 \\ 7 & 8 \end{vmatrix}$

D $\begin{vmatrix} -1 & -2 \\ 7 & 8 \end{vmatrix}$

5. Given that $4^{2-x} = 4(2^{1-x})$, the value of $x$ is

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-1$

D $1$

6. Given that $\ln(x+6) + \ln 9 = \ln(x+7) + \ln 6$, the value of $x$ is

A $-4$

B $4$

C $-3$

D $8$

7. If the roots of the quadratic equation $px^2 - qx - r = 0$ are real, then

A $q^2 - 4pr \geq 0$

B $q^2 + 4pr \geq 0$

C $q^2 + 4p > 0$

D $q^2 - 4pr > 0$

8. $\lim_{n\to\infty}\left(1 + \frac{2}{3n}\right)^n =$

A $e^{\frac{3}{2}}$

B $e^{\frac{2}{3}}$

C $e^2$

D $e^{\frac{1}{3}}$

9. Given two statements $p$ and $q$, the converse of $p \Rightarrow q$ is

A $p \Rightarrow \sim q$

B $q \Rightarrow p$

C $\sim p \Leftrightarrow q$

D $\sim q \Rightarrow p$

10. The ordered pair that satisfies the following simultaneous equations is

$$2x + 3y = 1,$$
$$3x - y = 7.$$

A (2, 1)

B (-2, -1)

C (1, 2)

D (2, -1)

11.

$$\lim_{x \to 0} \frac{1 - \cos x}{x} =$$

A 0
B -1
C $\infty$
D 1

12. Using the substitutions $u = 1 + x^2$

$$\int_0^1 2x(1 + x^2)^3 dx =$$

A $\frac{15}{4}$
B $\frac{7}{4}$
C 1
D $\frac{17}{4}$

13.

$$\int \frac{2x + 1}{2x} dx =$$

A $x + \frac{1}{2}\ln x + K$
B $1 + \frac{1}{2}x^{-1} + K$
C $x + 2\ln 2x + K$
D $x + 2\ln x + K$

14. Using De Moiver's theorem, the complex number $(\cos\theta + i\sin\theta)^5$ is equal to

A $(\cos\theta^5 + i\sin\theta^5)$
B $5(\sin\theta + i\sin\theta)$
C $(\cos 5\theta + i\sin 5\theta)$
D $\left(\cos\frac{\theta}{5} + i\sin\frac{\theta}{5}\right)$

15. Given that **u** and **v** are coplanar vectors, where $\mathbf{u} = 2\mathbf{i} - \mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = -3\mathbf{i} + 5\mathbf{j} + \mathbf{k}$, $\mathbf{u} \times \mathbf{v} =$

A $-31\mathbf{i} - 20\mathbf{j} + 7\mathbf{k}$
B $-31\mathbf{i} + 20\mathbf{j} + 7\mathbf{k}$
C $-31\mathbf{i} - 20\mathbf{j} - 7\mathbf{k}$
D $31\mathbf{i} + 20\mathbf{j} - 7\mathbf{k}$

16. A function $f(x)$ is continuous in the interval $[a, b]$ and differentiable on the interval $(a, b)$. The mean value theorem states that there exists a value $c$, where $a < c < b$, such that

A $f'(c) = \frac{f(b) + f(a)}{a - b}$
B $f'(c) = \frac{f(b) - f(a)}{b - a}$
C $f'(c) = \frac{f'(a) - f'(b)}{b - a}$
D $f'(c) \frac{f'(b) - f'(a)}{a - b}$

17. The tangent of the acute angle between the lines $y = 4x - 3$ and $y = x - 2$ is

A $-\frac{3}{5}$
B $-\frac{5}{3}$
C $\frac{3}{5}$
D $\frac{5}{3}$

18. The general solution of the differential equation $3\frac{dy}{dx} = \frac{1}{x^2} + x$

A $y = \frac{1}{3}\left(\frac{x^2}{2} - \frac{1}{x}\right) + c$
B $y = \frac{1}{3}\left(\frac{x^2}{2} + \frac{1}{x}\right) + c$
C $y = \frac{1}{3}\left(\frac{1}{x} - \frac{x}{2}\right) + c$
D $y = \frac{1}{3}\left(\frac{1}{x} + \frac{x}{2}\right) + c$

19. A geometric progression has first term 16 and sum to infinity 40. The common ratio of the progression is

A $\frac{2}{5}$
B $\frac{3}{5}$
C $\frac{1}{5}$
D $\frac{4}{5}$

20. The equation of a line is $\mathbf{r} = \mathbf{i} - 3\mathbf{j} + 7\mathbf{k} + \mu(3\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ and the equation of a plane is $2x + 3y - 2z = 5$. The sine of the acute angle between the line and the plane is

A $\frac{2}{7}$
B $\frac{3}{7}$
C $\frac{1}{7}$
D $\frac{5}{7}$

21. The solution set of $\frac{x+1}{x} \geq 0$

A $\{x : x \leq -1 or\ x \geq 0\}$
B $\{x: x \leq 0\ or x \geq 1\}$
C $\{x: -1 \leq x \leq 0\}$
D $\{x: x \geq -1\}$

22. Given that $4\cos\theta + 3\sin\theta \equiv R\sin(\theta + \alpha)$, the value of $R$ and $\tan\alpha$ are respectively

A 25 and $\frac{3}{4}$
B 5 and $\frac{3}{4}$
C 25 and $\frac{4}{3}$
D 5 and $\frac{4}{3}$

23. A circle with radius 4 has its centre at the point (2,4). The equation of the circle is

A $(x-2)^2 + (y-4)^2 = 16$
B $(x-2)^2 + (y-2)^2 = 4$
C $(x-4)^2 + (y-2)^2 = 16$
D $(x-2)^2 + (y-4)^2 = 4$

24. The general solution of the equation $\cos\theta = \frac{\sqrt{3}}{2}$ is

A $\theta = 2n\pi \pm \frac{\pi}{3}$
B $\theta = 2n\pi \pm \frac{\pi}{6}$
C $\theta = n\pi \pm \frac{\pi}{3}$
D $\theta = n\pi \pm \frac{\pi}{6}$

25. Given that $X$ and $Y$ are two variables related by the equation $Y = aX^2 + bX$, where $a$ and $b$ are constants. A suitable linear transformation for this non-linear equation is

A $\frac{Y}{X} = aX + b$
B $Y = X(aX + b)$
C $\ln Y = \ln X + \ln(aX + b)$
D $\ln Y = \ln(aX^2) + \ln(bX)$

26. A partial order relation is that which is

A Reflexive, symmetric and transitive
B Reflexive and symmetric
C Reflexive, anti symmetric and transitive
D Symmetric and transitive

27. Given the parametric equations $x = 2\theta + \sin 2\theta$, $y = 1 - \cos 2\theta$, where $\theta$ is the parameter, $\frac{dy}{dx} =$

A $\frac{\sin 2\theta}{1 - \cos 2\theta}$
B $\frac{\cos\theta}{\sin\theta}$
C $-\frac{\sin 2\theta}{1 + \cos 2\theta}$
D $\frac{\sin 2\theta}{1 + \cos 2\theta}$

28. Given that $(1 + ax)^{\frac{1}{2}} \equiv 1 + \frac{x}{2} + bx^2 + \cdots$,

A $a = -1, b = \frac{1}{4}$
B $a = 1, b = -\frac{1}{8}$
C $a = 1, b = -\frac{1}{4}$
D $a = -1, b = \frac{1}{8}$

29. The number of arrangements of the letter of the word *ASSASSINATION* is

A $\frac{13!}{3!\,4!\,2!}$

B $\frac{13!}{4!\,3!\,2!\,2!}$

C $\frac{5!}{4!\,3!\,4!}$

D $\frac{5!}{4!\,3!\,2!\,2!}$

30. Given that the sum of the first n terms of a sequence is $S_n$, where $S_n = n(n+2)$, the fourth term of the sequence is

A 24

B 15

C 11

D 9

31. The polynomial $p(x) = x^3 + ax^2 + 2x - 5$ leaves the remainder 2 when divided by $(x-1)$.The value of$a$ is

A 4

B -6

C -6

D 10

32. $\sum_{r=1}^{100}(r+1) =$

A 5050

B 200

C 10200

D 5150

33. A point P divides the line segment joining the points (4, 1) and (5, 2) internally in the ratio 2:3. The coordinates of P are

A $\left(\frac{22}{5}, 7\right)$

B $\left(\frac{22}{5}, \frac{7}{5}\right)$

C $\left(22, \frac{7}{5}\right)$

D $(22, 7)$

34. A periodic function $f$, is defined on $\mathbb{R}$ , the set of real numbers by

$$f(x) = \begin{cases} x+2, & 0 < x \le 3, \\ 2x - 13, & 3 < x \le 5, \end{cases}$$

If $f(x) = f(x+5k), k \in \mathbb{Z}, then\ f11 =)$

A 13

B 9

C 1

D 3

35. The value of the constant $m$ for which the graph ofthe equation $y = m(x-2)^2 + 3$passes through the origin is (0,0) is

A $\frac{3}{4}$

B $-\frac{3}{4}$

C $\frac{4}{3}$

D $-\frac{4}{3}$

## SECTION B: MECHANICS

36. At time $t = 0$ the position vectors of two particles $A$ and $B$ are $(5\mathbf{i} + 20\mathbf{j})$ m and $(-10\mathbf{i} - 5\mathbf{j})$ m respectively. The particles are given constant velocities $(-2\mathbf{i} - 5\mathbf{j})$ m s$^{-1}$ and $(3\mathbf{i} + 5\mathbf{j})$ m s$^{-1}$ respectively. The position vector of $B$ relative to the position vector of $A$ after 3 s is

A $5\mathbf{j}$ m

B $-5\mathbf{j}$ m

C $(-15\mathbf{i} - 25\mathbf{j})$ m

D $(5\mathbf{i} + 10\mathbf{j})$ m

37. Two particles of mass 2 kg and 3 kg are placed at the points (3, 1) and (8, 6) respectively. The position of their center of mass is at the point

A (4, 6)

B (5·5, 3·5)

C (5, 3)

D (6, 4)

38. A particle is projected with speed 20 m $s^{-1}$ at an angle 30° to the horizontal through the point of projection. Taking $g$ as 10 m $s^{-2}$, the greatest height reached by the particles from the point of projection is

A 25m
B 5 m
C $20\sqrt{3}$ m
D 10 m

39. A lorry of mass 1000 kg is moving with speed 3 m $s^{-1}$ when a 200 kg sack of rubbish is dropped vertically into the lorry. The new speed of the lorry is

A 12 m $s^{-1}$
B $2\frac{1}{2}$ m $s^{-1}$
C $2\frac{6}{7}$ m $s^{-1}$
D $\frac{3}{4}$ m $s^{-1}$

40. A body of weight 20 N is suspended from a fixed point $O$ by a light elastic string of natural length 0·3 m and modulus of elasticity 200 N, the extension of the string is

A 0·03 m
B 0·3 m
C 3 m
D 0.12m m

41. A particle is projected from a point on a horizontal plane with speed $V$ at an angle $\theta$ to the plane. The horizontal range $R$ covered by the particle is

A $R = \frac{2V^2 \sin 2\theta}{g}$
B $R = \frac{V^2 \cos\theta \sin\theta}{g}$
C $R = \frac{V^2 \sin 2\theta}{2g}$
D $R = \frac{V^2 \sin 2\theta}{g}$

42. Two forces **R**= (2**i** + 3**j**) N and **F** = (8**i** + 7**j**) N act on a body of mass 10 kg. The velocity of the body 4 s after starting from rest is

A (24**i** + 16**j**)m $s^{-1}$
B (4·0 **i** + 4·0 **j**) m $s^{-1}$
C (2·4**i** + 1·6**j**) m $s^{-1}$
D (40**i** + 40**j**) m $s^{-1}$

43. A force of 14 N acts on a particle of mass 70 kg causing the speed of the particle to increase from 3 m $s^{-1}$ to 7 m $s^{-1}$. The distance travelled by the particle during this period is

A 40 m
B 4 m
C 100 m
D 10 m

44. Fig. 1 shows two particles $P$ and $Q$ hanging freely and connectedby a light inextensible string passing over a fixed smooth pulley.

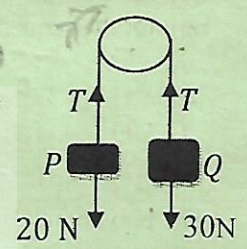


Fig. 1

The system is released from rest. Taking $g$ as 10 m $s^{-2}$, the acceleration of $P$ and the tensionin the string are

A 10 m $s^{-2}$ and 24 N
B 2 m $s^{-2}$ and 20 N
C 10 m $s^{-2}$ and 30 N
D 2 m $s^{-2}$ and 24 N

45. A particle of mass 4 kg is moving in horizontal circle of radius 2 m. The centripetal force is 72 N. The angular speed of the particle is

A 9 rad $s^{-1}$
B 3 rad $s^{-1}$
C 6 $ms^{-1}$
D 3 m $s^{-1}$

46. The position vector of a particle is
$\mathbf{r} = [(t^3 + 2t^2 + 2)\mathbf{i} + (t^2 - 1)\mathbf{j}]$ m
at time $t$ s. The velocity of the particle when $t = 1$ is

A (9**i** + **j**)m $s^{-1}$
B (7**i** + 2**j**)m $s^{-1}$
C (7**i** + **j**) m $s^{-1}$
D 5im $s^{-1}$

47. The speeds of a car during 8 s of braking are given in the table below.

| Time, $t$ s | 0 | 2 | 4 | 6 | 8 |
|---|---|---|---|---|---|
| Speed, $v$ m s$^{-1}$ | 30 | 27 | 22 | 14 | 0 |

Using the trapezium rule, the distance covered by the car before coming to rest is approximately

A 156 m
B 158 m
C 157 m
D 155 m

48. The engine of a car of mass 1200 kg works at a constant rate of 16 kW. The car is moving on a straight level road with constant resistance of magnitude 400 N to its motion. The maximum speed of the car is.

A 40 m s$^{-1}$
B 4 m s$^{-1}$
C 13 m s$^{-1}$
D 80m s$^{-1}$

49. A fair die is thrown two times in succession. The probability of throwing at least *one* six is

A $\frac{1}{36}$
B $\frac{12}{36}$
C $\frac{11}{36}$
D $\frac{2}{6}$

50. The work done in stretching an elastic string of natural length 5 m and modulus of elasticity 500 N from a total length of 6 m to 7 m is

A 1300 J
B 400 J
C 150 J
D 300 J

**STOP**

**GO BACK AND CHECK YOUR WORK**